Fusion Output Analysis

OCR Model Performance

Comparison of character error rate (CER), word error rate (WER), and character similarity across model groups.

Group	Models/Combination	CER %	WER %	Similarity %
G1	LlamaIndex, Hunyuan, Surya	15.90	18.04	76.63
G2	Hunyuan, Chandra, Llamaindex	5.65	6.11	92.25
G3	Docling, Llamaindex, Hunyuan	5.38	6.75	88.11
G4	Llamaindex, hunyuan, Deepseek	5.22	6.97	85.6
G5	Docling , Tesseract , Easyocr	25.91	31.71	71.74
G6	DeepSeekocr2, Sonnet4, Docling	8.52	8.05	78.67
G7	Docling, Deepseek, Hunyuan	11.63	11.70	77.23
G8	Chandra, qwen25, Sonnet4	9.20	9,47	77.40
G9	DeepSeekocr2, Lightonocr, Mistral	44.08	50.82	33.48
G10	GLM, Mistral, Hunyuan	48.61	57.66	27.81
G11	Mistral, DeepSeek, Sonnet4	43,72	50,09	32.63
G12	Docling, Hunyuan, Chandra	11.32	11.32	76
G13	Chandra, Lightonocr, Surya	30.00	33.22	44.76
G14	GLM, Doctr , Easyocr	60.56	77.63	70.37
G15	Qianfan , Chandra , PaddleOCR	21.38	23.70	81.48
G16	Gemini2.5,Llamaparse,pdfplumbe	35.60	34.66	48.89

Interpretation rules:

Lower CER/WER = better accuracy
Higher similarity = better semantic consistency

Analysis

The evaluation examines sixteen groups of merged OCR models, each constructed from a distinct combination of components selected for their complementary strengths and the ways in which their errors tend to correlate. The purpose of this hybrid fusion approach is to generate outputs that closely resemble the original document, ensuring that the combined models reinforce one another and effectively mitigate individual weaknesses.

Group 1, which consists of Hunyuan, LlamaIndex and Surya, shows a mediocre performance. With a CER of 15.90% and a WER of 18.04%, the group has the highest error rate in the evaluation. The similarity value of 76.63% is also the lowest. This suggests that Surya in this configuration does not harmonize well with Hunyuan and LlamaIndex, and that the error correlations between the models reinforce each other rather than compensate. The group therefore has difficulty both transcribing correctly and preserving the semantic structure of the text.

Group 2, which also consists of Hunyuan, LlamaIndex and Chandra, is the most semantically accurate group. With a similarity value of 92.25**%,** it preserves the meaning and structure of the text better than all other groups. In addition, its CER of 5.65**%** and WER of 6.11**%** shows that the group maintains a high level of accuracy. The combination of the models suggests that Chandra contributes strong semantic stability, while Hunyuan and LlamaIndex stand for robust character and word recognition. This makes group 2 particularly suitable for tasks where meaning preservation is more important than absolute character accuracy.

Group 3, consisting of Docling, Hunyuan and LlamaIndex, is in a stable middle layer. With a CER of 5.38%, WER of 6.75%, and a similarity of 88.11%, the group shows a smooth and reliable performance. Docling's contribution seems to provide some variation in error patterns, improving semantic consistency. The group is balanced but not exceptional.

Group 4, DeepSeek, LlamaIndex and Hunyuan, shows relatively good accuracy at the character and word level. With a CER of 5.22% and a WER of 6.97%, the group is one of the most precise in the evaluation. However, the similarity value of 85.62% is lower than in group 2 and group 3. This suggests that DeepSeek contributes to high detail precision but not as strong semantic stability.

Group 5, Docling, Tesseract, and EasyOCR, demonstrates comparatively lower accuracy among the evaluated combinations. With a CER of 25.91% and a WER of 31.71%, the group exhibits significantly higher error rates than the top-performing groups. Its similarity score of 71.74% indicates a moderate ability to preserve the structure and meaning of the text. The results suggest that while Docling contributes positively to recognition quality, the inclusion of Tesseract and EasyOCR introduces higher error variability. Consequently, this group serves as a useful baseline but is less suitable for applications requiring high precision and strong semantic fidelity.

Group 6, DeepSeekOCR2, Sonnet4, and Docling, demonstrates solid mid-tier performance. With a CER of 8.52% and a WER of 8.05%, the group maintains reasonable transcription accuracy. Its similarity score of 78.67% indicates moderate semantic preservation. This suggests that while the models cooperate effectively in reducing recognition errors, they do not achieve the semantic robustness observed in higher-performing groups such as Group 2. Nevertheless, the combination provides a dependable balance between accuracy and contextual integrity.

Group 7,Docling, DeepSeek, and Hunyuan, shows moderate performance across all metrics. Recording a CER of 11.63%, a WER of 11.70%, and a similarity score of 77.23%, the group exhibits acceptable but not outstanding results. The relatively aligned CER and WER values indicate consistent recognition behavior, though the semantic fidelity remains limited. This configuration may be suitable for applications where moderate accuracy is sufficient.

Group 8, Chandra, Qwen2.5, and Sonnet4, achieves a balanced performance. With a CER of 9.20%, a WER of 9.47%, and a similarity score of 77.40%, the group maintains steady accuracy and reasonable semantic consistency. Chandra's contribution appears to enhance semantic stability, while Qwen2.5 and Sonnet4 support reliable transcription. Overall, the group offers a well-rounded solution without excelling in any single metric.

Group 9, DeepSeekOCR2, LightOnOCR, and Mistral, demonstrate poor performance. With a CER of 44.08% and a WER of 50.82%, the group exhibits extremely high error rates. Its similarity score of 33.48% further indicates substantial loss of semantic integrity. These results suggest significant incompatibilities among the models, making the group unsuitable for tasks requiring reliable text recognition or meaning preservation.

Group 10, GLM, Mistral, and Hunyuan, delivers the weakest semantic performance in the evaluation. With a CER of 48.61% and a WER of 57.66%, it shows the highest transcription errors, accompanied by a very low similarity score of 27.81%. This indicates severe degradation of both textual accuracy and semantic structure. The combination appears highly ineffective and unsuitable for practical OCR or transcription applications.

Group 11, Mistral, DeepSeek, and Sonnet4, also performs poorly. With a CER of 43.72% and a WER of 50.09%, the group demonstrates substantial transcription inaccuracies. Its similarity score of 32.63% confirms limited semantic preservation. The results suggest that the interaction between these models amplifies recognition errors rather than mitigating them.

Group 12, Docling, Hunyuan, and Chandra, presents moderate and stable performance. With identical CER and WER values of 11.32% and a similarity score of 76.00%, the group shows balanced but unexceptional results. The inclusion of Chandra contributes to semantic coherence, while Docling and Hunyuan provide structural consistency. This combination may be considered reliable but not optimal.

Group 13, Chandra, LightOnOCR, and Surya, exhibits relatively weak performance. With a CER of 30.00% and a WER of 33.22%, the group experiences substantial transcription errors. Its similarity score of 44.76% indicates limited semantic accuracy. These findings suggest that the models do not effectively complement one another.

Group 14, GLM, Doctr, and EasyOCR, demonstrates mixed results. Despite extremely high error rates—60.56% CER and 77.63% WER—the similarity score of 70.37% is comparatively high. This discrepancy suggests that, although transcription accuracy is poor,

the models preserve certain structural or contextual aspects of the text. Nonetheless, the group remains unsuitable for high-precision applications.

Group 15, Qianfan, Chandra, and PaddleOCR, delivers relatively strong performance. With a CER of 21.38% and a WER of 23.70%, the group achieves moderate accuracy, while its similarity score of 81.48% indicates good semantic preservation. Chandra's contribution appears to enhance contextual fidelity, making this group suitable for tasks prioritizing meaning over exact character accuracy.

Group 16, Gemini 2.5, LlamaParse, and pdfplumber, shows below-average performance. With a CER of 35.60% and a WER of 34.66%, the group demonstrates notable transcription errors. Its similarity score of 48.89% indicates limited semantic consistency. This suggests that document parsing tools, when combined with language models, may introduce structural inconsistencies that affect recognition accuracy.

Conclusion

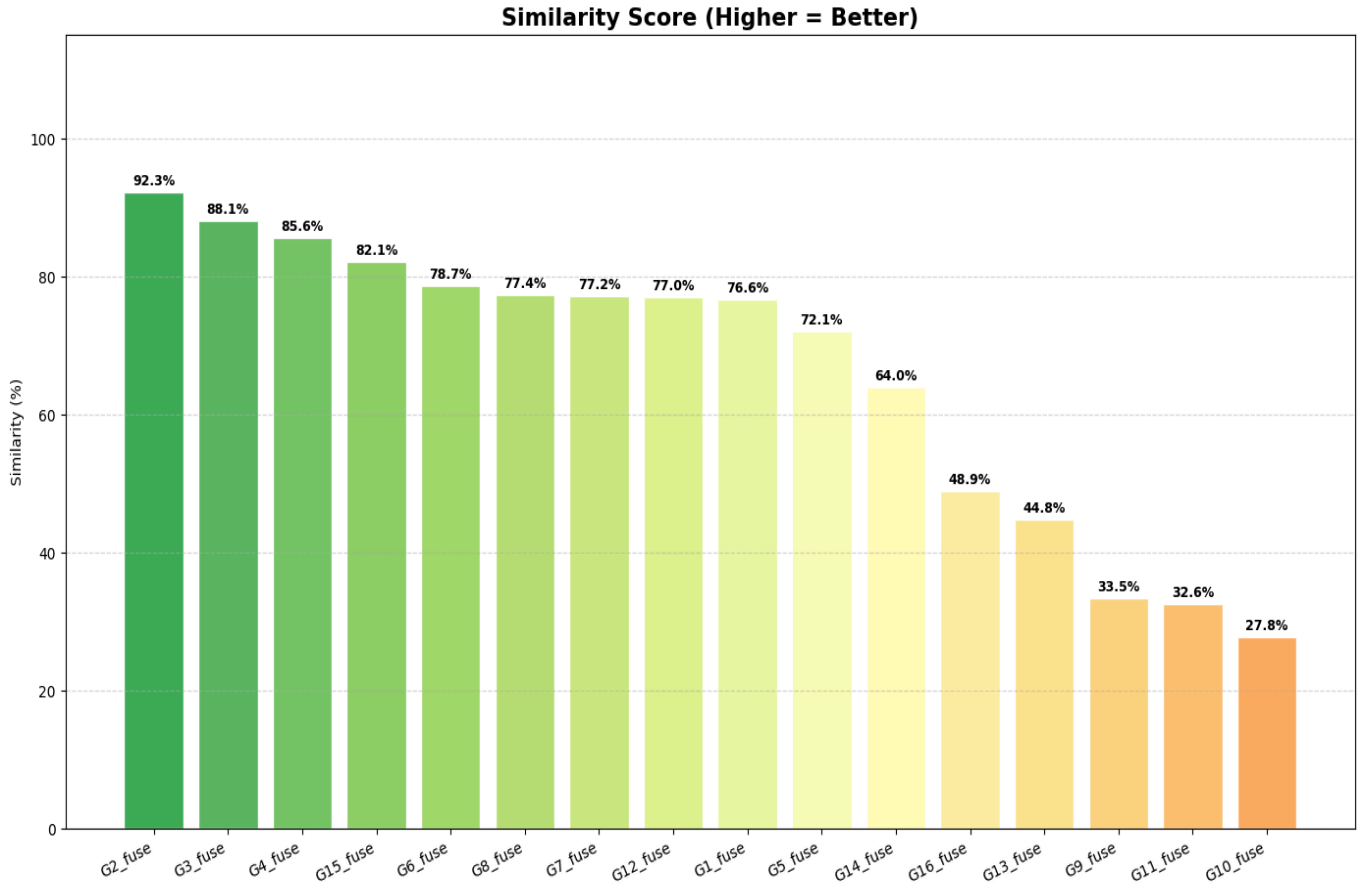
The evaluation clearly shows that Group **2** delivers the strongest overall performance. Its combination of Hunyuan, LlamaIndex, and Chandra achieves the best semantic similarity while maintaining low error rates. This suggests that the models' error patterns are complementary and that Chandra contributes meaningful semantic reinforcement.

Group **4** excels in raw accuracy, likely due to DeepSeek's contribution to fine-grained recognition. However, its lower similarity score indicates that the models do not fully compensate for each other's semantic limitations.

Group **3** remains a dependable and balanced configuration. Docling introduces beneficial diversity in error behavior, but the synergy is not strong enough to surpass the top groups.

Lowerperforming groups—particularly those with CER above 10%—likely suffer from overlapping weaknesses, where similar error tendencies accumulate rather than cancel out. This leads to both high transcription errors and reduced semantic fidelity.

Overall, the results reinforce a key principle of hybrid OCR pipelines:
Model combinations with complementary strengths and low error correlation consistently outperform those with overlapping weaknesses.


Similarity Score (Higher = Better)
Similarity (%)
0
20
40
60
80
100
92.3%
88.1%
85.6%
82.1%
78.7%
77.4%
77.2%
77.0%
76.6%
72.1%
64.0%
48.9%
44.8%
33.5%
32.6%
27.8%
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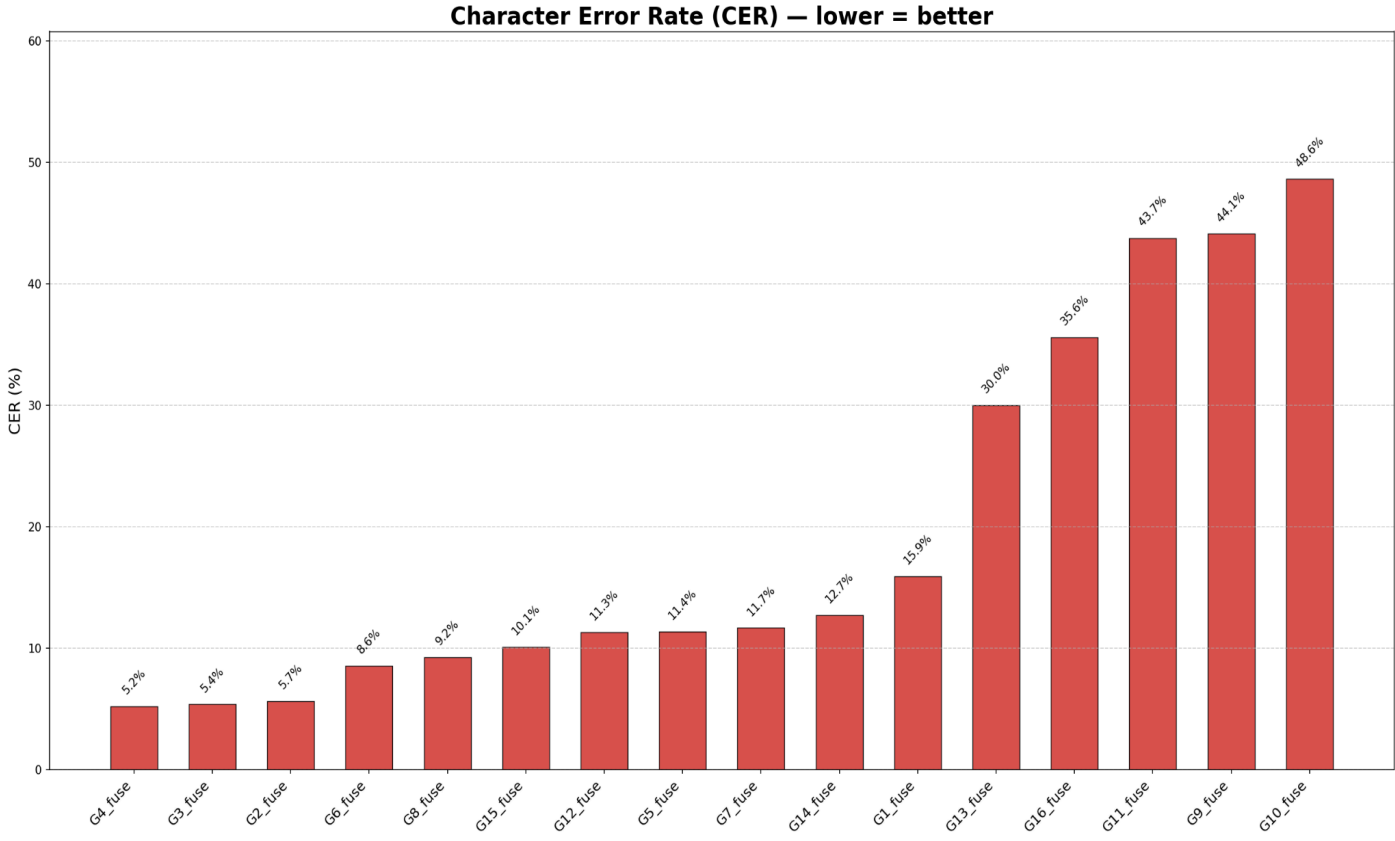
Character Error Rate (CER) — lower = better
CER (%)
0
10
20
30
40
50
60
5.2%
5.4%
5.7%
8.6%
9.2%
10.1%
11.3%
11.4%
11.7%
12.7%
15.9%
30.0%
35.6%
43.7%
44.1%
48.6%
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G2_fuse
G6_fuse
G8_fuse
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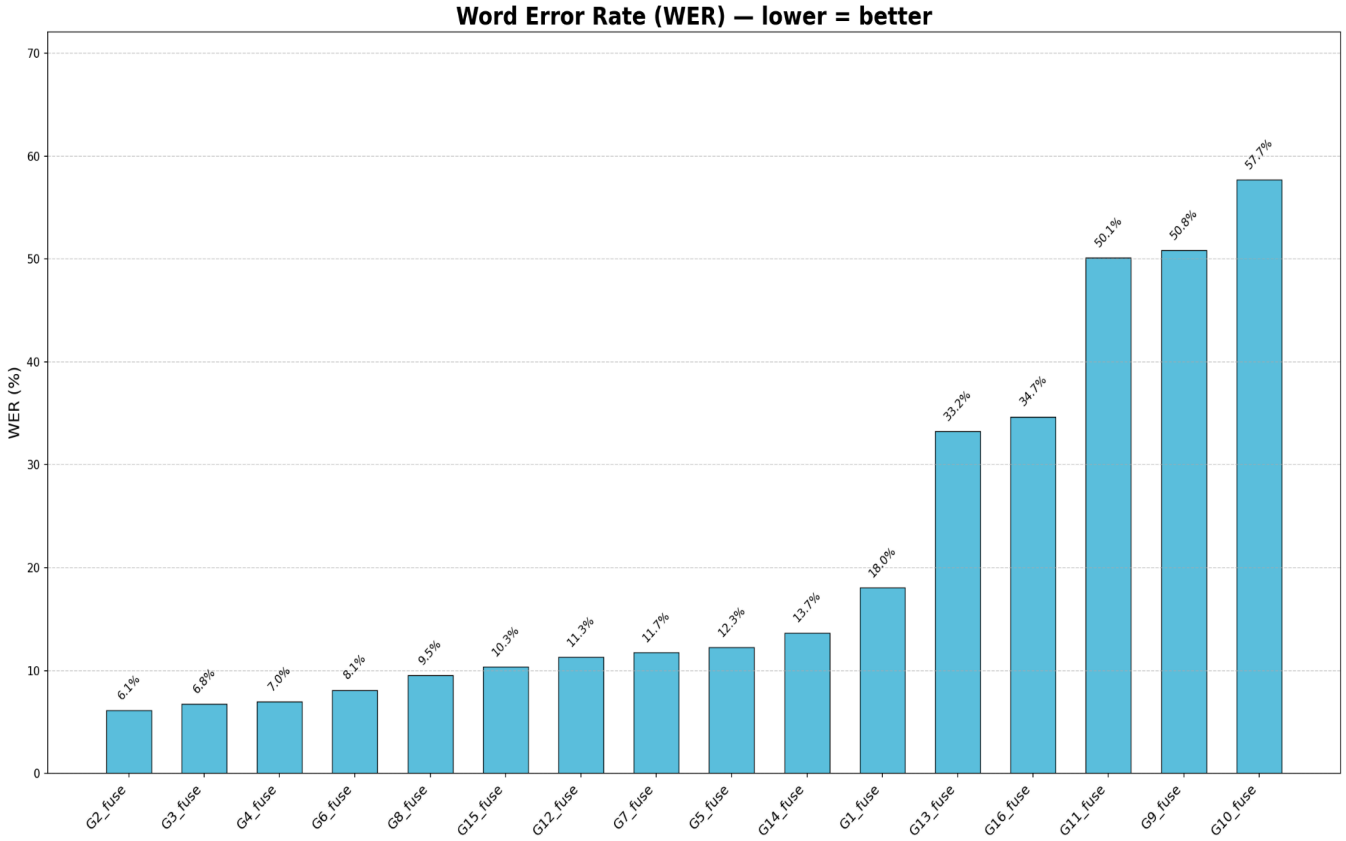
Word Error Rate (WER) — lower = better
WER (%)
0
10
20
30
40
50
60
70
6.1%
6.8%
7.0%
8.1%
9.5%
10.3%
11.3%
11.7%
12.3%
13.7%
18.0%
33.2%
34.7%
50.1%
50.8%
57.7%
G2_fuse
G3_fuse
G4_fuse
G6_fuse
G8_fuse
G15_fuse
G12_fuse
G7_fuse
G5_fuse
G14_fuse
G1_fuse
G13_fuse
G16_fuse
G11_fuse
G9_fuse
G10_fuse